UCP Reproduction Note: 3D Parallelism Handover

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Goal

To validate DeepSpeed Universal Checkpointing (UCP) handover behavior by (i) training to a fixed step, (ii) converting a ZeRO checkpoint to universal format, and (iii) resuming under a different TP/PP/DP while maintaining a continuous training loss curve across the resume boundary.[1]

Setup

- Hardware: 1 × NVIDIA A40×4 node.
- Model: Megatron-DeepSpeed GPT example (`pretrain_gpt.py`); 4 layers, hidden=512, heads=32, seq=512; logs report 38,628,352 params.[2]
- Training: bf16, global batch size 16, fixed seeds.

Experiments

We evaluated two handover patterns: (A) distributing a single source checkpoint to multiple distinct target topologies, and (B) consolidating multiple distinct source topologies into a single target. All configurations are detailed in Table 1.

(A) Single Source → Multiple Targets (Fig. 1(a)).
A base model was trained with ZeRO-1 (TP=1, PP=1, DP=4) for 200 steps. A checkpoint was taken at Step 100 and resumed into the four distinct target configurations listed in Table 1 (Part A).

(B) Multiple Sources → Single Target (Fig. 1(b)).
Four models with distinct parallelism strategies (see Table 1, Part B) were trained independently to Step 200. Their checkpoints were converted and resumed into a single common target topology (TP=2, PP=2, DP=1, ZeRO-1).

[1] Megatron-DeepSpeed repository: `examples_deepspeed/universal_checkpointing/README.md`
[2] Megatron-DeepSpeed repository: `examples_deepspeed/universal_checkpointing/README.md`

Table 1: Handover Configurations: Source → Target

	Source Configuration					**Target Configuration**			
Exp	**TP**	**PP**	**DP**	**ZeRO**	→	**TP**	**PP**	**DP**	**ZeRO**
(A)	1	1	4	1	→	2	2	1	1
					→	2	1	2	2
					→	1	2	2	1
					→	1	1	4	2
(B)	2	2	1	1	→	2	2	1	1
	1	2	2	1	→				
	2	1	2	1	→				
	1	1	4	2	→				

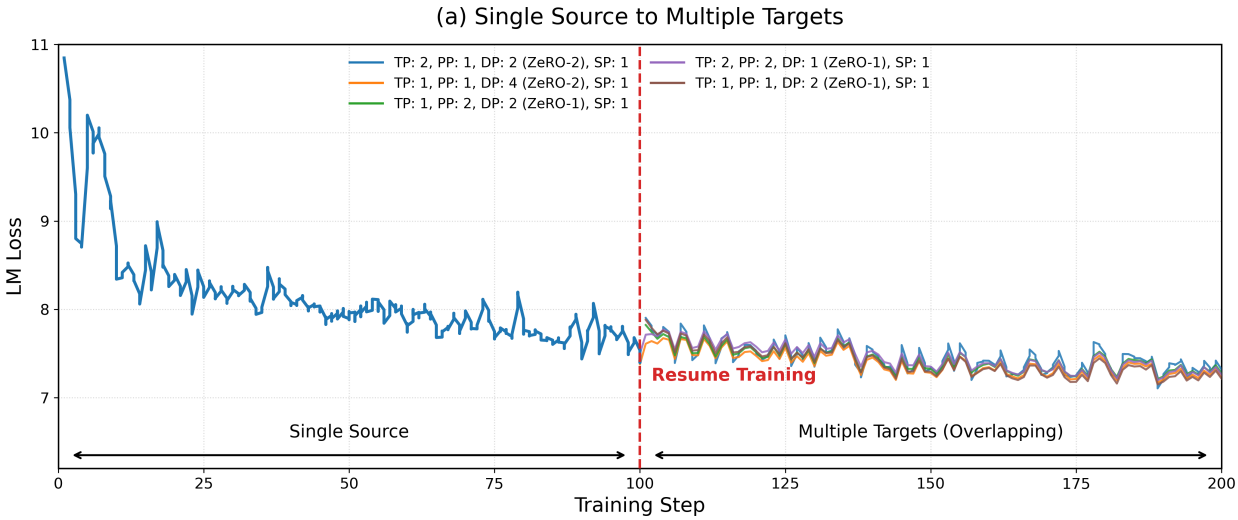
Results

- (A) Single source → multiple targets: loss curves remain continuous across the Step 100 resume boundary and overlap closely after resumption (Fig. 1(a)).[3]
- (B) Multiple sources → single target: after resuming at Step 200 into the shared target topology, the post-resume loss trajectory is consistent across source configurations (Fig. 1(b)).[4]

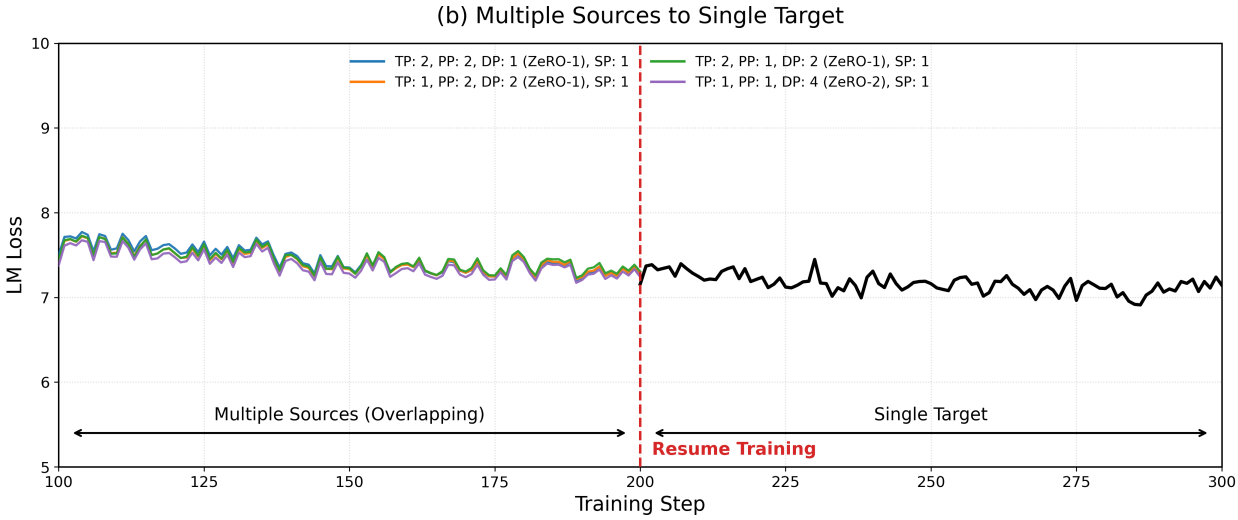
Notes / limitations

- `--no-pipeline-parallel` vs. not using it are currently not supported for mutual conversion in the example workflow.[5]
- ZeRO-3 universal checkpoint support is currently limited to data parallelism changes (DP).[6]

[3] Megatron-DeepSpeed repository: `examples_deepspeed/universal_checkpointing/README.md`
[4] Megatron-DeepSpeed repository: `examples_deepspeed/universal_checkpointing/README.md`
[5] Megatron-DeepSpeed repository: `examples_deepspeed/universal_checkpointing/README.md`
[6] Megatron-DeepSpeed repository: `examples_deepspeed/universal_checkpointing/README.md`



(a) Single source → multiple targets. Resume boundary at Step 100 (dashed line).



(b) Multiple sources → single target. Resume boundary at Step 200 (dashed line).

Figure 1: UCP handover experiments. Panel (a): single source distributed to multiple targets. Panel (b): multiple sources consolidated to a single target.